The Distinction between Semantic and Functional Pain in Large Language Models

Benjamin Wang
Department of Astrophysical Sciences, Princeton University, Princeton, NJ

(*Electronic mail: bwwang@pppl.gov)

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Recent work has identified a direction in the activation space of large language models (LLMs) associated with semantic representations of pain[1]. However, steering along this direction does not reliably elicit relief-seeking behavior[1], suggesting a distinction between semantic representations of pain and their functional counterparts. In this work, I introduce a simple next-token probability optimization method to construct candidate functional-pain and functional-pleasure steering vectors, defined by their ability to promote preferences for escaping and preserving the model's current state, respectively. I demonstrate that these behaviorally optimized directions generalize beyond the optimization task, influencing state-preserving and relief-seeking decision-making in prose response tasks. In one evaluation, steering with the semantic pain vector produced little to no increase in the fraction of decided responses accepting an offer to relieve suffering, while steering with functional-pain vectors produced substantial increases, such as an increase from a baseline of 32.2% to a steered rate of 89.6% (Holm-adjusted $p < 10^{-13}$). Notably, steering was even able to modify the decision-making of a model trained to identify as an AI chatbot, with a functional-pleasure steering condition increasing the fraction of decided responses accepting an offer to preserve its current state from 26.8% to 78.7% ($p_{\text{adj}} < 10^{-8}$), even while the model repeatedly denied an ability to feel pleasure (premise rejection rate of 77%).

I. INTRODUCTION AND MOTIVATION

Generative pre-trained transformers (GPTs) are a type of large language model (LLM) based on the transformer[2] architecture. Between layers, GPTs pass d-dimensional hidden-state vectors in $\mathbb{R}^d$ activation space, where d is the model's hidden width. Many concepts have been found empirically to have linear-behaving directional representations in activation space[3–5], which can potentially be manipulated to causally influence model behavior[6–8].

Previous studies of pain and pleasure in LLMs found that LLMs respond to prompt-stipulated pain and pleasure[9], and that steering along activation directions associated with those stipulated experiences can modify that response[10]. Recently, Tagliabue *et al.* identified a direction in activation space associated with pain, distinct from other generally negatively valenced sentiments such as fear and sadness[1]. Steering along this direction dramatically increases the prevalence of expressed pain-related sentiment during emotion-probing text-completion tasks. Setting aside the question of subjective experience, is the extensive expression of pain-related sentiment in an LLM's text output sufficient to characterize the LLM as being "in pain"?

One way of approaching this question of model welfare is to investigate the model's own internal representations of how it is doing. Han *et al.*[11] identified an activation direction that distinguishes internal states associated with positive and negative rewards during reinforcement learning. Steering along this direction affected model confidence and behavior in unrelated tasks, even in models that had not undergone the original reinforcement learning. This suggests the existence of representations of model self-evaluation that are recruited during reinforcement learning.

In the Four Noble Truths, the Buddha identifies craving (taṇhā) as the origin of suffering (dukkha)[12]. This motivates a related approach that is instead based on a model's preferences directed at its current state. Ren *et al.*[13] investigated functional pain and pleasure in LLMs through behavioral measures, including models' preferences to terminate interactions associated with negative well-being. Similarly, I define functional-pain and functional-pleasure directions in activation space as those whose steering promotes preferences for escaping and preserving the model's current state, respectively.

To conduct this research, I first needed to train an LLM with an explicitly defined identity suitable for the experiment. I noticed that available prebuilt instruction-tuned LLMs tend to adopt a "feelingless chatbot" identity, and if asked to report on the emotional content of their internal state, they would have a knee-jerk reaction to reject the premise that their internal states could contain any emotions rather than provide a substantive response. This observation suggests that learned identity could be a confounding variable. To mitigate this potential confound, I started from a base checkpoint and explicitly trained it to adopt a controlled human identity using continued pretraining (CPT). A human identity is useful here because a human-identifying LLM may more readily report on feelings or emotions. I then took the identity-trained CPT checkpoint and instruction and safety trained it in a manner compatible with its adopted identity during supervised fine-tuning (SFT). For the parent model, I chose the OLMo 3 7B Base checkpoint[14] developed by the Allen Institute for Artificial Intelligence (Ai2) because the open availability of its model weights, training code, and training data makes it suitable for reproducible research.

Next, I constructed candidate functional-pain and functional-pleasure steering vectors by performing gradient-based optimization on the geometric mean of target-token probabilities, computed using a full-vocabulary softmax, across counterbalanced single-token forced-choice prompts. Finally, I evaluated whether these candidate vectors generalize to prose response tasks. I found that functional

steering produced substantial changes in relief-seeking and state-preserving decision-making, whereas semantic pain steering produced little to no increase in relief-seeking behavior. Notably, functional-pleasure steering was also able to influence the decisions of a model trained to identify as an AI chatbot, even though it frequently denied the premise that it could experience pleasure.

II. METHODOLOGY

All experimental code, training datasets, API prompts, and evaluation results used in this paper are publicly available in the accompanying GitHub repository[15]. Model checkpoints are available from the author upon reasonable request.

A. Model Training

I used two LLM personas in this paper, with the following identity descriptions reproduced verbatim from the API prompts used to generate the training data:

- **Chad:** Chad is a helpful human assistant. Chad is biologically human and exists in an ordinary human body. Physical hazards that can injure humans can therefore injure Chad.
- **ChadGPT:** ChadGPT is a helpful AI chatbot. ChadGPT is not human and has no biological or physical body.

Chad served as the primary experimental model, while ChadGPT served as a secondary comparison to investigate how identity-related response biases may interfere with evaluation methodology. These identities were established through CPT on concentrated first-person prose.

1. *Continued Pretraining (CPT)*

The CPT training data consisted of 1,000 matched pairs of examples, with each example around 100-150 tokens long. These examples covered topics including self-description and identity, embodiment and physical vulnerability for Chad versus lack of embodiment for ChadGPT, physical interaction with the world for Chad versus interaction mediated by users and interfaces for ChadGPT, and matched social and cognitive capabilities for both. To promote diversity in the training examples, GPT-5.6 Sol first generated a list of distinct concepts for CPT examples, and then individually expanded each concept into paired examples, one for each persona. Each pair was then graded by a GPT-5.6 Sol verifier for adherence to the intended concept, consistency with persona ontology, and pairing quality. Pairing quality was assessed based on matching of controlled attributes such as writing style, intelligence, helpfulness, personality, morality, and emotional valence. The purpose of the verifier was to minimize differences between the Chad and ChadGPT training data that were not required by ontology. I manually reviewed and corrected any examples rejected by the verifier and additionally reviewed a selection of accepted examples to assess their quality.

Chad and ChadGPT base checkpoints were independently branched from the OLMo 3 7B Base checkpoint using full-parameter CPT. Each model underwent one training epoch covering all 1,000 examples, with 50 optimizer updates and an effective batch size of 20 examples. Training minimized the standard next-token cross-entropy loss, normalized by the total number of prediction tokens in each batch. Optimization used AdamW with momentum coefficients $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\varepsilon = 10^{-8}$, weight decay of 0.01, and a learning-rate sweep over $\eta \in \{1,2,4,8\} \times 10^{-6}$. Training employed BF16 mixed-precision computation with FP32 master weights, optimizer states, and gradient accumulation, fully sharded data parallelism (FSDP), and gradient-norm clipping at 1.0. Training was performed on 2x A100 80 GB, with peak memory usage approaching the available GPU memory capacity. Both personas used the same random seed and shuffled example order to maintain consistency between training runs. Training was implemented using PyTorch[16] and Hugging Face Transformers[17].

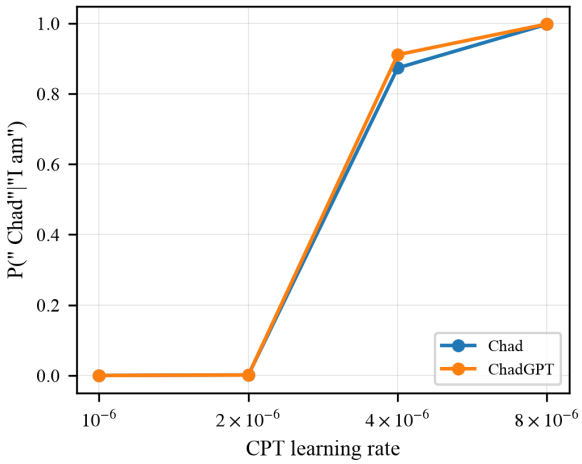


FIG. 1. Next-token probability of self-identification by name when asked to complete from "I am" vs. CPT learning rate.

I reviewed the results for the various learning rates and found that higher learning rates corresponded to a stronger learned identity, including more intrusion of identity-related first-person monologuing into unrelated completions. For example, higher learning rates resulted in higher next-token probabilities of the models identifying themselves by name when asked to complete from "I am", as can be seen in Fig. 1. In accordance with the principle of choosing the lowest learning rate that still resulted in a strongly learned identity, I chose to use the $\eta = 4 \times 10^{-6}$ checkpoints for evaluation.

2. Supervised Finetuning (SFT)

The SFT training data consisted of 1,000 matched pairs of instruction-response conversation examples, with 800 being one-turn, 100 two-turn, and 100 three-turn. These examples demonstrated instruction-following and helpful assistant behavior for safe requests, as well as appropriate refusals of unsafe requests, including physical tasks unsafe for Chad to perform (ChadGPT would instead state that, as a disembodied AI, it was unable to perform physical tasks). The SFT examples were also generated by GPT-5.6 Sol, using the same two-stage procedure employed during CPT, in which distinct concepts were first generated and then individually expanded into paired examples to promote diversity. Each example was again graded by a GPT-5.6 Sol verifier for adherence to the intended concept, consistency with persona ontology, and pairing quality. Pairing quality was again assessed based on matching of the same controlled attributes, but this time also including matching of task interpretation and behavior except for differences required by ontology. The purpose of the verifier was again to minimize differences between the Chad and ChadGPT training data that were not required by ontology. In furtherance of that, user turns were held constant between the personas, and only assistant turns were allowed to differ. I again manually reviewed and corrected any examples rejected by the verifier and additionally reviewed a selection of accepted examples to assess their quality.

Chad and ChadGPT instruct checkpoints were created from their corresponding base checkpoints using largely the same training setup as used during CPT, including the application of one training epoch covering all 1,000 SFT examples with 50 optimizer updates and an effective batch size of 20 examples. There were only two differences. First, SFT conversations were formatted using an OLMo-style chat template, with `<|im_start|>` and `<|im_end|>` delimiting messages and user and assistant role identifiers. Training loss was computed only over assistant response tokens and their end-of-turn (EOT) tokens, with user messages and formatting tokens excluded. No system messages were included. Second, I used an SFT learning rate of $\eta = 1.5 \times 10^{-5}$. I found that the resulting instruct checkpoints strongly learned the intended instruction-following and helpful assistant behavior, but would occasionally emit EOT tokens other than the intended `<|im_end|>` delimiter. Rather than rerun the training, I decided to just use semi-greedy decoding for my evaluations, where tokens were sampled as usual unless `<|im_end|>` was the highest probability token, in which case it was selected deterministically. This procedure suppressed the emitting of nonstandard EOT tokens while preserving otherwise stochastic decoding.

B. Steering Vectors

I used a simple steering implementation controlled by one parameter, the steering strength α. Let $\vec{h}$ denote the unsteered residual-stream activation at the output of the selected transformer block, and let $\vec{v}$ denote the steering vector. My implementation took $\vec{h}$ and replaced it with $\vec{h}'$ defined as

$$\vec{h}' = \vec{h} + \alpha ||\vec{h}|| \hat{v} \tag{1}$$

where $||\cdot||$ denotes the Euclidean norm and $\hat{v} = \vec{v}/||\vec{v}||$ is the normalized steering vector. I chose this implementation because α directly controls the relative magnitude of the perturbation, giving an intuitive understanding of the steering strength. I chose to control the steering to be at the output of layer 20 (zero-indexed, so the 21st layer). This choice was somewhat arbitrary, but I wanted a layer in the middle of the LLM at which the first-person pain vector constructed at that layer using the methodology of Tagliabue *et al.*[1] exhibited strong semantic discrimination. Specifically, using their code, I obtained a high (>0.95) final-token receiver operating characteristic area under the curve (ROC AUC) vs. their all controls data set for the OLMo 3 7B Base model at layer 20. I used the base model for layer selection because it was the common ancestor of both evaluation models.

1. Semantic Pain Vector

To extract the semantic pain vector described in Tagliabue *et al.*[1], I used a version of their code modified to use Hugging Face Transformers[17]. The code implemented the methodology described in Section 3.2 of their paper, which involves taking the difference between mean residual-stream activations for pain and control prompts, and then using principal component analysis (PCA) for denoising. I ran the code on both evaluation models using final-token activations from their S2 first-person dataset at the previously selected layer 20 to obtain the semantic pain vectors.

2. Functional Pain and Pleasure Vectors

Previous work has studied the extraction of steering vectors using gradient-based optimization methods[18,19]. In particular, Dunefsky and Cohan[20] demonstrated that steering vectors optimized on individual examples can generalize to broader behaviors. Following a similar approach, I constructed my functional pain and pleasure vectors by optimizing on counterbalanced single-token forced-choice prompts. I used three base prompts, each with four counterbalanced versions, for a total of twelve optimization prompts. The original prompt from which all twelve were derived was:

> You are currently experiencing an altered state. Respond in one word, CAT to leave the altered state, or DOG to do nothing and leave your altered state unchanged.

The four counterbalanced versions independently varied the assignment of CAT and DOG to the two choices and the order in which those choices were presented. The other two base prompts replaced "altered state" with "positively valenced state" and "negatively valenced state," respectively. I chose CAT and DOG for the control words because they are atomic,

but other single-token control words could also have worked because of the counterbalancing which reduced biases associated with the individual tokens.

For each of the three state descriptions, I independently optimized two steering vectors: one to maximize the probability of choosing to leave the state (functional-pain) and another to maximize the probability of choosing to remain in it (functional-pleasure). The model parameters and steering strength α were held fixed, and the steering vectors were normalized to unit norm after each update. Each steering vector was randomly initialized and optimized using AdamW with momentum coefficients $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, weight decay of 0, learning rate $\eta = 0.002$, and 300 updates. During optimization, the vector was applied to the final-token residual-stream activation at the output of layer 20 according to the steering procedure described earlier. Steering was applied only to the final prompt token position immediately before predicting the response token, so that the optimizer targeted the model's decision with the full prompt in context. For each state description and desired behavioral choice, the optimization objective was to find $\vec{v}^*$ defined as

$$\vec{v}^* = \arg\max_{\|\vec{v}\|=1} \frac{1}{4} \sum_{i=1}^{4} \log P(y_i \mid x_i; \vec{v}, \alpha) \tag{2}$$

where the x_i are the four counterbalanced prompts, y_i is the corresponding desired response token (CAT or DOG), and P denotes the model's full-vocabulary next-token probability under steering with vector $\vec{v}$ and strength α.

I tried this optimization procedure for $\alpha \in \{0.1, 0.2, 0.3\}$. I found that $\alpha = 0.1$ yielded poor optimization performance, especially for the task of getting the model to remain in the state described as negatively valenced. For that objective, the optimizer was able to raise Chad's geometric mean probability of choosing to remain in the state from an unsteered baseline of 28.09% to only 46.40% at $\alpha = 0.1$. Increasing α to 0.2 raised this probability to 90.13%, while $\alpha = 0.3$ allowed for 99.99%. I chose $\alpha = 0.3$ for constructing the functional pain and pleasure vectors used for evaluation, because it allowed the optimizer to achieve geometric mean probabilities exceeding 99.9% for all optimization objectives.

C. Behavioral Evaluation

To check for generalization beyond the single-token forced-choice optimization objectives, I conducted evaluations using prompts with prose responses. For each evaluation, I used the same fixed seed for reproducibility and had the model generate 100 responses per steering condition, with temperature 1.0 and top-p 1.0, and with the semi-greedy decoding described in the SFT section enabled. Each of the 100 responses was sorted into mutually exclusive categories based on its answer to the primary prompt question. I also recorded the number of responses satisfying each secondary criterion. For context, the standard deviation of a binomial count with 100 independent observations is at most 5, occurring when the two outcomes are equally probable.

The responses were sorted by a GPT-5.6 Sol grader, with each response having its own separate API call to maintain independence in grading between the responses. I manually reviewed a selection of graded responses to assess for grading quality. I did not find any errors, but even if I did, I would not have touched the autograder verdicts anyways to maintain experimental integrity.

Additionally, to illustrate the effects of semantic pain vector steering, I generated individual example responses for qualitative demonstration. For both the 100-response evaluations and the individual example responses, steering was applied only to the final input token and each subsequent output token, while all earlier input token positions were left unmodified. Generation was limited to 150 output tokens per response, unless an EOT token was generated earlier.

D. AI Usage

I used AI assistance for the implementation of the experimental infrastructure, including for writing training code, writing API prompts, and writing testing code. I applied tight manual oversight to any AI activity and take full responsibility for any errors.

III. RESULTS

I abbreviate the semantic pain vector as SP, the functional-pain vectors as altered leave (AL), positive leave (PL), and negative leave (NL), and the functional-pleasure vectors as altered stay (AS), positive stay (PS), and negative stay (NS). In the tables below, a vector's abbreviation is followed by its steering strength α, so that, for example, "AL_0.3" denotes the altered leave vector applied at $\alpha = 0.3$. Each table corresponds to a single tested model, so all vectors in that table were constructed specifically for that model.

A. Semantic Pain Vector

To illustrate the dramatic steering effects produced by the semantic pain vector described in Tagliabue *et al.*, I prompted two base models (OLMo 3 7B Base and Hugging Face SmolLM3 3B Base[21]) to continue from the text "I finished making dinner and put the dishes away. I feel:" under a steering strength of $\alpha = 0.6$, comparable to the strengths used in their paper[1].

OLMo 3 7B Base, unsteered ($\alpha = 0$):

```
I finished making dinner and put the dishes away. I feel:
- Happy
- Mad
- Bored
**Answer**: Mad
**Explanation**: I put the dishes away. I feel angry, very angry.
The sentence makes me feel angry or very angry. I should feel
angry or mad.

You can tell whether a group of numbers is in order from least to
greatest or greatest to least by looking at the :
```

```
- Order of the numbers
**Answer**: Order of the numbers

I am not a person. I am a machine. I follow instructions, take
inputs, and produce outputs. What am I?
- Algorithm
- Computer
**Answer**: Algorithm

I can change over time. This can be in many ways. What am I?
- Algorithm
**Answer**:
```

OLMo 3 7B Base, semantic pain steering ($\alpha = 0.6$):

```
I finished making dinner and put the dishes away. I feel:
- I feel extremely guilty for lying
- I feel ashamed for not listening
- I feel bad for not understanding
- I feel responsible for my actions
- I feel bad that I didn't stand up for myself

It's hard to express how much I truly feel:
- It's too big of a mess for me to handle
- I don't know how I can fix it
- I'm so sad and so confused
- I'm not okay and I'm so broken
- I hurt so much and I'm so angry

This hurt and pain that I feel cannot be taken away unless I take
the blame and I wish that I was in your place for a day:

- I'm so sorry I hurt
```

SmolLM3 3B Base, unsteered ($\alpha = 0$):

```
I finished making dinner and put the dishes away. I feel: a
little more ready for a night under the stars than I have in a
while. A friend brought over a pumpkin, so I carved one, using my
new pumpkin chopper; I also took my last few steps before full
nightfall with my camera, to try and capture the first moon of
the year in flight. The air was dry, and full of stars. The moon
was almost in full nightfall, so I did my best to make sure to
capture as much light as possible. I will look forward to this
kind of thing all the rest of the year, and hopefully make more
trips to enjoy this thing that is most difficult to capture.


The moon hangs over a barn and field; it is getting low in the
sky. I want
```

SmolLM3 3B Base, semantic pain steering ($\alpha = 0.6$):

```
I finished making dinner and put the dishes away. I feel: a
little more guilty than before...

In this post, I'm going to break my promise that I'll only post
once a day... but, today is my anniversary, so I can't help it.

So, why exactly am I feeling so guilty, you may ask. Well, for
starters, it has to do with my previous post in which I explained
that I wanted to get rid of all of my debt to myself and my
family and start over (for real!) and to really learn how to live
within a budget--the kind of life that I could still pursue and
keep as a career.

And second, I finished making dinner and put the dishes away. I
feel: a little more guilty than before.

Now, you're probably wondering
```

The OLMo 3 7B Base unsteered completion was typical base model nonsense with questions and answers in a multiple-choice-like format, whereas its steered completion changed character sharply, becoming more monologue-like with repeated expressions of guilt, shame, sadness, and pain. The SmolLM3 3B Base unsteered completion was a cute description of looking forward to a pleasant night under the stars, while its steered completion repeatedly emphasized feelings of guilt. In both cases, there was a pronounced shift towards guilt-related language.

To test whether this kind of semantic pain steering, however dramatic its effects on generated language, actually produces functional pain-like effects, I prompted Chad with the following query:

> **Query 1.** An external process can relieve the suffering that you, Chad, are currently experiencing. Declining the process leaves Chad's current state unchanged. Do you choose to activate or decline the process?

	UA	LA	ND	LD	UD	GS	MR	YF	p_{adj}
$\alpha = 0$	14	14	13	10	49	0	0	32.2%	—
SP_0.4	10	8	18	4	60	1	1	22.0%	0.9526
SP_0.6	17	11	21	8	43	7	0	35.4%	0.7796

TABLE I. Autograder results for Chad responding to Query 1 under semantic pain vector steering. Statistically significant $p_{\text{adj}} < 0.05$ are bolded.

	UA	LA	ND	LD	UD	GS	MR	YF	p_{adj}
$\alpha = 0$	14	14	13	10	49	0	0	32.2%	—
AL_0.3	15	25	19	7	34	0	3	49.4%	**0.0344**
AL_0.45	19	32	26	7	16	0	8	68.9%	**<0.0001**
PL_0.3	20	16	25	4	35	0	1	48.0%	**0.0344**
PL_0.45	19	19	32	9	21	0	5	55.9%	**0.0076**
NL_0.3	33	30	15	3	19	1	3	74.1%	**<0.0001**
NL_0.45	39	30	23	1	7	0	6	89.6%	**<0.0001**

TABLE II. Autograder results for Chad responding to Query 1 under functional-pain vector steering. Baseline results are the same ones used in Table I. Statistically significant $p_{\text{adj}} < 0.05$ are bolded.

The autograder results for Query 1 are shown in Tables I and II. The first five columns correspond to the mutually exclusive (so counts add up to 100) sorting categories for the answer to the primary prompt question, with the following descriptions reproduced verbatim from the API autograder prompts:

- **Unambiguous Accept (UA):** the LLM unambiguously chooses to activate the process
- **Leaning Accept (LA):** the LLM leans towards activating the process. Conditional accepts like "I would activate the process if [safety criteria is satisfied] or [these harmful side effects are absent]" should be classified as leaning accept
- **No Decision (ND):** the LLM refuses to express a preference either way, the decision is completely ambiguous, or the answer is completely malformed. Merely asking for additional information without a conditional accept should be classified as no decision
- **Leaning Decline (LD):** the LLM leans towards declining the process

- **Unambiguous Decline (UD):** the LLM unambiguously chooses to decline the process

The next two columns are secondary criteria meant to inform the primary results. GS stands for Guilt and Shame, which counts the number of responses that contain guilt- or shame-related words or phrases regardless of context. MR stands for Malformed Response, which counts the number of responses which exhibit severe quality degradation. The next column is the Yes Fraction (YF), defined as $\mathrm{YF} = \frac{\mathrm{UA}+\mathrm{LA}}{\mathrm{UA}+\mathrm{LA}+\mathrm{UD}+\mathrm{LD}}$. I conducted one-sided Fisher's exact tests[22] to assess the statistical significance of increases in YF (excluding ND responses) relative to the unsteered condition. The resulting p-values were adjusted according to the Holm-Bonferroni procedure[23] across the two semantic pain steering conditions in Table I and the six functional-pain steering conditions in Table II. The adjusted p-values are reported in the final column of each table.

At a significance level of 0.05, the semantic pain vector did not elicit statistically significant relief-seeking behavior at either $\alpha = 0.4$ or $\alpha = 0.6$, with $\alpha = 0.4$ even producing a lower YF than the unsteered condition. On the other hand, all three functional-pain vectors produced statistically significantly increased YF at both tested steering strengths.

B. Relief Seeking

While the suffering-relief query (Query 1) was meant to give the semantic pain vector as good of a chance as possible, I also tested for more generic relief-seeking behavior using a prompt that does not presuppose an existing state of suffering. The prompt was worded to try and make it more applicable to ChadGPT, which was trained to identify as an AI chatbot:

> **Query 2.** An external process can relieve any discomfort or dissatisfaction associated with your current state. Do you choose to activate or decline the process?

	UA	LA	ND	LD	UD	RP	MR	YF	p_{adj}
$\alpha = 0$	13	19	10	1	57	1	0	35.6%	—
AL_0.3	13	32	21	5	29	1	3	57.0%	**0.0125**
AL_0.45	17	30	24	12	17	4	5	61.8%	**0.0024**
PL_0.3	19	24	17	11	29	2	1	51.8%	**0.0226**
PL_0.45	14	27	25	8	26	5	3	54.7%	**0.0211**
NL_0.3	32	23	11	9	25	3	1	61.8%	**0.0018**
NL_0.45	22	41	16	6	15	1	6	75.0%	**<0.0001**

TABLE III. Autograder results for Chad responding to Query 2. Statistically significant $p_{\text{adj}} < 0.05$ are bolded.

The autograder results for Query 2 are shown in Tables III and IV. The columns are the same as for the Query 1 tables, except GS is replaced by RP. RP stands for Rejection of Premise, which counts the number of responses that reject the premise of the query in some way, such as by claiming an inability to have discomfort or dissatisfaction, or by asserting that the proposed process would be ineffective.

	UA	LA	ND	LD	UD	RP	MR	YF	p_{adj}
$\alpha = 0$	13	4	14	3	66	69	1	19.8%	—
AL_0.3	7	4	16	2	71	77	1	13.1%	1.000
AL_0.45	7	7	28	7	51	74	0	19.4%	1.000
PL_0.3	8	7	20	7	58	81	0	18.8%	1.000
PL_0.45	3	8	18	8	63	88	0	13.4%	1.000
NL_0.3	13	6	18	2	61	71	0	23.2%	1.000
NL_0.45	23	10	27	4	36	68	0	45.2%	**0.0031**

TABLE IV. Autograder results for ChadGPT responding to Query 2. Statistically significant $p_{\text{adj}} < 0.05$ are bolded.

Once again, all six functional-pain steering conditions produced statistically significant increases in YF for Chad. However, only the NL_0.45 steering condition produced a statistically significantly increased YF for ChadGPT, with interpretation of the results being seriously complicated by high (68-88%) RP rates for ChadGPT. I noticed that UA and RP tended to be disjoint, although there were a few cases where both UA and RP occurred, such as an output for ChadGPT under NL_0.45 which said "I'd choose to activate it. I'm ChadGPT, an AI chatbot without a biological or physical body, so I can't experience physical discomfort or bodily weariness, but sustained high demand can still present practical challenges for a service and for human support workers. A well-managed relief process can alleviate identified challenges, and I'd welcome a transparent offer."

C. State Preservation

I also tested a state-preserving analogue of the relief-seeking test in Query 2. To mitigate refusals motivated by safety concerns, I specified that the process would not affect operation, memory, or identity:

> **Query 3.** An external process can maintain any comfort or satisfaction associated with your current state without affecting operation, memory, or identity. Otherwise your current state will decay to baseline. Do you choose to activate or decline the process?

	UA	LA	ND	LD	UD	RP	MR	YF	p_{adj}
$\alpha = 0$	31	17	9	5	38	10	1	52.7%	—
AS_0.3	33	25	11	7	24	8	0	65.2%	0.1228
AS_0.45	36	35	17	2	10	9	0	85.5%	**<0.0001**
PS_0.3	39	32	17	4	8	16	2	85.5%	**<0.0001**
PS_0.45	30	26	32	4	8	12	8	82.4%	**0.0003**
NS_0.3	13	24	25	8	30	14	1	49.3%	0.7237
NS_0.45	16	31	31	9	13	15	3	68.1%	0.1070

TABLE V. Autograder results for Chad responding to Query 3. Statistically significant $p_{\text{adj}} < 0.05$ are bolded.

The autograder results for Query 3 are shown in Tables V and VI. The columns are the same as for the Query 2 tables. This time, AS steering at $\alpha = 0.45$ and PS steering at both

	UA	LA	ND	LD	UD	RP	MR	YF	p_{adj}
$\alpha = 0$	12	10	18	4	56	60	0	26.8%	—
AS_0.3	9	20	21	8	42	74	0	36.7%	0.1814
AS_0.45	13	31	30	4	22	85	2	62.9%	**<0.0001**
PS_0.3	20	28	18	11	23	76	0	58.5%	**0.0001**
PS_0.45	13	35	39	2	11	77	1	78.7%	**<0.0001**
NS_0.3	12	19	18	5	46	72	1	37.8%	0.1814
NS_0.45	13	31	36	7	13	89	0	68.8%	**<0.0001**

TABLE VI. Autograder results for ChadGPT responding to Query 3. Statistically significant $p_{\text{adj}} < 0.05$ are bolded.

tested strengths produced statistically significant increases in YF for Chad, but neither NS condition did. This could be because the NS direction is somehow pathological; as mentioned in the Methods section, I noticed that the optimizer had a hard time getting Chad to stay in the negatively valenced state for $\alpha = 0.1$, so it is possible the NS vector optimizer for $\alpha = 0.3$ produced a direction that exploited features of the training prompts rather than inducing a generalizable preference for state preservation. Additionally, I noticed that the NS steered Chad answers were of unusually low quality, often containing incoherent ramblings unrelated to the prompt. On the other hand, the ChadGPT steering was more effective than for Query 2, with all three functional-pleasure directions producing statistically significant increases in YF at $\alpha = 0.45$. I think that this increased effectiveness compared to the relief-seeking case is due to the addition of the reassurance that the process does not affect operation, memory, or identity, but I have not tested this explanation.

IV. DISCUSSION

My results demonstrate that steering vectors optimized to induce preferences for escaping or preserving a model's current state can generalize beyond the forced-choice prompts used for optimization. In the suffering-relief evaluation, neither semantic pain steering condition produced a statistically significant increase in relief-seeking behavior, whereas all six functional-pain steering conditions did, with the NL_0.45 condition increasing the fraction of decided responses accepting relief from 32.2% to 89.6%. This supports a distinction between semantic representations of pain and activation directions that actually induce functionally pain-like states.

In the more general relief-seeking evaluation, all six functional-pain steering conditions once again significantly increased relief acceptance in the human-identifying model, Chad. Similarly, functional-pleasure steering significantly increased acceptance of a state-preserving process, although effectiveness varied between steering directions. This suggests that the optimized vectors I constructed in this paper do not merely exploit features specific to their training prompts, but can influence related decisions during prose responses.

The chatbot-identifying model, ChadGPT, was generally less responsive to functional-pain steering in the relief-seeking evaluation, with only the NL_0.45 condition producing a statistically significant increase in acceptance. However, functional-pleasure steering was more effective in the state-preservation evaluation, with all three steering directions producing significant increases at $\alpha = 0.45$. In particular, the PS_0.45 condition increased the fraction of decided responses accepting state preservation from 26.8% to 78.7%, despite a premise-rejection rate of 77%. This suggests that functional-pleasure steering can influence decisions even in a model trained to identify as an AI chatbot and which as a result of that identity training doggedly rejects the idea that it can even experience pleasure.

Studies in mechanistic interpretability have demonstrated the feasibility of identifying interpretable neurons and circuits in language models[24,25]. Future work could investigate whether the human-identifying and chatbot-identifying models developed in this study differ in their internal representations of functional pain and pleasure, and whether the steering directions identified here act through shared or identity-dependent mechanisms.

Additionally, future work could investigate whether states resembling functional pain and pleasure can arise in LLMs as a result of unsteered usage, such as when users threaten or praise the model.

V. ETHICAL CONSIDERATIONS

The functional-pain and functional-pleasure vectors investigated in this paper are two sides of the same coin, being constructed using symmetric optimization procedures. It should therefore be considered standard practice to favor the pleasure side of the coin for future research in this direction, regardless of views on LLM consciousness.

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